

Here is the complete instruction framework and architectural logic to build an automated agent that evaluates LinkedIn profiles, computes an ICP score (0-100) using embeddings, and manages company data caching.



THE AUTOMATED ICP SCORING FRAMEWORK

This framework is divided into 4 operational phases: **Data Ingestion & Caching**, **Semantic Embedding**, **Structural Validation**, and **Score Computation**.

PHASE 1: Data Ingestion & Caching (The "Fetch" Layer)

Objective: Extract minimal profile data and resolve company headcount using a cache to save API/scraping costs.

Instruction Rules for the Agent:

- 1. Extract Profile Payload:** From the target LinkedIn profile, extract *only* the "Minimal Semantic Payload":
 - * `profile_preview_text`: The text summary/preview near the full name.
 - * `profile_description`: The "About" section or headline text.
 - * `current_job_title`: The exact current job title.
 - * `company_name`: The name of the current employer.
- 2. Cache Check (Company Headcount):**
 - * Query the local database/cache using `company_name` as the key.
 - * **IF Cache Hit:** Retrieve `cached_employee_count`.
 - * **IF Cache Miss:** * Navigate to the Company LinkedIn Page. * Extract the employee count range (e.g., "11-50 employés", "51-200 employés"). * Save `{ "company_name": "X", "employee_count": "11-50 employés" }` to the cache. * Set `current_employee_count` to this value.

PHASE 2: Semantic Embedding & Comparison (The "Brain" Layer)

Objective: Compare the "fewest relevant data" (the Minimal Semantic Payload) against the ICPs using vector embeddings.

Instruction Rules for the Agent:

- 1. Prepare the ICP Vectors:** Pre-compute and store the embedding vectors for the core semantic parts of the ICPs (Titles + Keywords).
 - * **Vector ICP 1:** Embeds ("Fondateur", "Co-fondateur", "Gérant", "Managing Partner", "Directeur d'agence", "Directeur des Opérations", "cabinet de recrutement", "chasseur de tête", "conseil en recrutement", "croissance", "IA", "automatisation", "RGPD", "conformité").
 - * **Vector ICP 2:** Embeds ("Associé", "Partner", "Fondateur", "Directeur de cabinet", "Directeur Général", "cabinet de chasse", "executive search", "recherche de cadres", "headhunting", "cabinet de recrutement spécialisé", "nous recrutons", "développement", "digital").
- 2. Prepare the Profile Vector:** Combine the `profile_preview_text`, `profile_description`, and `current_job_title` into a single string. Generate its embedding vector

(Vector Profile). 3. **Calculate Cosine Similarity:** * Compute similarity between Vector Profile and Vector ICP 1 -> Score_Semantic_1 (0.0 to 1.0). * Compute similarity between Vector Profile and Vector ICP 2 -> Score_Semantic_2 (0.0 to 1.0). * **Max Semantic Score:** $\text{Max_Semantic} = \max(\text{Score_Semantic_1}, \text{Score_Semantic_2})$.

PHASE 3: Structural Validation (The "Gatekeeper" Layer)

Objective: Embeddings are bad at exact numbers. We must apply hard logic rules for Location and Headcount.

Instruction Rules for the Agent: 1. **Location Check:** Does the profile location or company location contain "France", "Paris", "Lyon", etc.? (Boolean: True/False). 2. **Headcount Match:** Compare current_employee_count against ICP criteria (11-50 for ICP1, 1-50 for ICP2). * *Exact Match:* True (e.g., profile is 11-50, ICP1 is 11-50). * *Adjacent Match:* Partial (e.g., profile is 51-200, ICP1 is 11-50). * *No Match:* False (e.g., profile is 200+).

PHASE 4: ICP Score Computation (0-100)

Objective: Synthesize semantic similarity and structural rules into a single 0-100 score.

The Formula: $\text{Total ICP Score} = (\text{Semantic Weight}) + (\text{Headcount Weight}) + (\text{Location/Industry Weight})$

- **1. Semantic Weight (Max 60 points):**

- $\text{Max_Semantic} * 60$. (If cosine similarity is 0.85, they get 51 points. This rewards profiles that naturally use the right vocabulary and context).

- **2. Headcount Weight (Max 25 points):**

- Exact Match to ICP1 or ICP2 range = **25 points**.
- Adjacent Match (e.g., 51-200 when targeting 11-50) = **10 points**.
- No Match (e.g., 1000+ employees) = **0 points**.

- **3. Location & Industry Weight (Max 15 points):**

- Location is France AND Industry is HR/Recruitment = **15 points**.
- Location is France BUT Industry is wrong (or vice versa) = **5 points**.
- Neither = **0 points**.

(Note: If the profile is a 100% semantic match but has 10,000 employees, the headcount drops them to a 60/100, correctly identifying them as a bad fit despite looking like the right persona).



OUTPUT: THE PROPER LINKEDIN PROFILE

ANALYSIS TEMPLATE

When the agent finishes processing, it must output the following structured analysis card. This is the final deliverable for the user/system.

LINKEDIN PROFILE ICP ANALYSIS REPORT

1. PROFILE IDENTITY & EXTRACTED DATA

- Full Name: [Name]
- Current Title: [Exact Title]
- Company Name: [Company Name]
- Profile Preview/Description Summary: [1-2 sentence summary of the extracted text near the name]
- Minimal Semantic Payload Extracted: [List the exact keywords/phrases pulled for embedding]

2. COMPANY CONTEXT & CACHE STATUS

- Company Name: [Company Name]
- Employee Count: [e.g., 11-50 employés]
- Cache Status: [HIT / MISS & UPDATED]
- Industry Detected: [e.g., Services aux ressources humaines]
- Location Detected: [e.g., Paris, France]

3. SEMANTIC EMBEDDING ANALYSIS

- Vector ICP 1 (Scaling Cabinet) Similarity: [e.g., 0.82 / 1.0]
- Vector ICP 2 (Niche/Exec Search) Similarity: [e.g., 0.45 / 1.0]
- Primary ICP Match: [ICP 1 or ICP 2]
- Semantic Alignment Notes: [e.g., "Profile heavily aligns with ICP1 keywords: 'automatisation', 'cabinet de recrutement', 'RGPD'."]

4. STRUCTURAL VALIDATION (HARD FILTERS)

- Location Match (France): [✅ YES / ❌ NO]
- Industry Match (HR/Recruitment): [✅ YES / ❌ NO]
- Headcount Match (1-50 or 11-50): [✅ EXACT / ⚠️ ADJACENT / ❌ NO MATCH]

5. FINAL ICP SCORE & VERDICT

- Semantic Score: [XX] / 60
- Headcount Score: [XX] / 25
- Location/Industry Score: [XX] / 15

🏆 TOTAL ICP SCORE: [XX / 100]



ACTIONABLE VERDICT:

- [80-100] HOT: Immediate outreach. Highly aligned semantically and structurally.
- [60-79] WARM: Good fit, but [state the missing point, e.g., "headcount is slightly too large"]. Proceed with cautious outreach.
- [40-59] COLD: Lacks semantic alignment or structural fit. Nurture or ignore.
- [<40] REJECT: Not an ICP match. Discard.



AGENT RECOMMENDATION:

[1 sentence generated by the LLM on the best angle to approach this specific person based on the semantic payload. E.g., "Since they mentioned 'RGPD' and 'automatisation' in their preview, lead the outreach with the compliance-by-design angle of the Recruteur Augmenté offer."]

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Implementation Rules for your Automation Tool (Make/n8n/Python):

1. **Embedding Model:** Use a multilingual model like `text-embedding-3-small` (OpenAI) or `multilingual-e5-large` (HuggingFace) because the profile text and ICPs are in French.
2. **Cache Database:** Use a simple key-value store (Redis, Supabase, or even a local JSON/SQLite file) mapping `company_name` -> `employee_count`.
3. **Prompting the LLM for the final output:** Pass the structured data (Scores, Cache status, Payload) to an LLM with a system prompt instructing it to fill out the "AGENT RECOMMENDATION" and format the text exactly as the template above. Do not let the LLM calculate the math; calculate the math in code, and only use the LLM for the qualitative summary.